

1 Executive Summary

Demand forecasting and warehouse-layout planning are critical pain points in logistics, with errors costing companies **trillions** of dollars annually. Global retailers lose ~\$1.2–1.75 trillion per year from stockouts and excess inventory. Forecast error rates in supply chains often run 20–40%, and less than half of businesses have confidence in their forecasts. Inaccurate forecasts lead directly to lost sales and wasted costs (each 4% sales lost from stockouts, inventory carrying costs ~20–30% of inventory value). Likewise, suboptimal warehouse storage and slotting inflate labor and picking costs: warehouse labor typically accounts for **60–65%** of fulfillment costs, and each picking error drains ~11–13% of profitability.

Classical solutions (statistical models, ML for forecasting; mixed-integer programming and heuristics for slotting) have improved results but scale poorly with complexity. For example, McKinsey found that supply-chain leaders achieve ~75–80% forecast accuracy, beyond which further gains yield little benefit. Forecasting models still struggle with volatile, fragmented data and demand shifts, while warehouse slotting problems grow combinatorially with SKUs and locations. Common KPIs – forecast accuracy (MAPE), fill rate, on-time delivery, inventory turns, labor productivity, and throughput – remain suboptimal under current methods. Implementation of best-practice systems (like modern WMS) can cut costs ~10–15%, but much larger gains seem elusive.

Quantum computing (QC) is emerging as a potential accelerator. QC's ability to explore many solutions in parallel offers hope for these hard problems. Early evidence suggests quantum approaches *could* yield orders-of-magnitude speedups on some optimization tasks. For instance, D-Wave pilots report *simulated* 10× reductions in rehandling by optimizing warehouse flow. However, QC is still experimental: current devices (annealers with thousands of qubits, gate systems with tens of qubits) can tackle only small proofs of concept. Major companies (DHL, Maersk, Volkswagen, Toyota, etc.) are already piloting quantum supply-chain projects, but broad enterprise adoption awaits more mature hardware.

This report examines **demand forecasting** and **warehouse storage optimization** in detail: quantifying the scale of pain, current methods and limits, key KPIs, and economic stakes. We then propose an *integrated demand–inventory optimization* problem that jointly plans forecasts and warehouse allocation. Finally, we compare classical vs quantum approaches (algorithms, complexity, maturity, data needs) in a summary table, and provide recommended metrics and interview questions for further exploration.

1.1 Demand Forecasting: Pain Points and Impacts

Forecasting future demand is notoriously difficult across industries (retail, e-commerce, manufacturing, 3PL). Key causes of error include volatile consumer behavior (seasonality, promotions, trends), fragmented data (silos between sales, marketing, supply), and lagging planning cycles. Gartner reports forecast errors of **20–40%** in retail and consumer goods chains. In practice, less than 50% of sales leaders trust their forecasts, and small shifts in demand can still trigger large upstream inventory swings (the “bullwhip” effect).

Classical ML and statistical methods (ARIMA, exponential smoothing, tree ensembles, deep neural nets) are widely used, but have limitations. They often require large clean historical datasets and compute time for training. They may not capture sudden market changes or rare events well. In large retail chains, models must be trained for thousands of SKUs and regions, making scenario analysis (e.g. Monte Carlo demand sampling) extremely compute-intensive. Even when highly accurate, forecasts are delivered with latency – studies note that forecasts often arrive too late to prevent misalignments.

KPIs and economic impact: Poor forecasts degrade many KPIs. Service level (fill rate) and customer satisfaction drop with stockouts, while inventory carrying costs and obsolescence rise with overstocks. Harvard Business Review estimated that stockouts cause ~4% of a retailer's sales to be lost. McKinsey found AI-based forecasting can cut product unavailability by up to 65%, recapturing much lost revenue. IHL Group data puts global inventory distortion (out-ages+excess) cost at roughly **\$1.77 trillion**, with stockouts alone ~ \$1.2T. A 15% increase in forecast accuracy is linked to ~3% higher pre-tax profit, and a 10–20% accuracy gain can yield 2–3% more revenue. Real cases back this: one manufacturer saved ~\$9M/year by improving demand planning. These magnitudes show why enterprises invest heavily in demand planning (top firms achieve ~10–15% better accuracy than peers) – yet reach a plateau around 75–80% accuracy where marginal returns drop.

Computational bottlenecks: Generating probabilistic forecasts or simulating “what-if” scenarios for thousands of products is compute-heavy. ML training can take hours on GPU clusters, and retraining as new data arrives is slow. Complex models (deep nets, ensemble forecasters) may require non-convex optimization. These bottlenecks mean forecasts often lag real demand shifts. Scalability is also an issue: small and mid-sized firms may lack the data or compute to use advanced ML, whereas large global carriers must run models at huge scale (millions of orders), where even a 1–2% efficiency gain can save **millions of dollars**.

Summary of challenges: Demand forecasting failures (rooted in data latency, model limitations, and volatility) lead to downstream issues: stockouts, emergency freight, markdowns, and distrust in planning. Forecasting KPIs (MAPE, bias, service level) remain stubbornly far from perfect, with each point of error carrying large cost.

1.2 Warehouse Storage & Slotting Optimization: Pain Points and Impacts

Warehouse storage optimization – deciding where and how to place inventory to minimize handling and travel – is another major headache. In large fulfillment centers, thousands of SKUs and orders flow each day. Inefficient layouts or slotting rules multiply handling: for example, in gravity-flow rack systems a misplaced item forces many “re-insertions” of other SKUs, causing wasted labor. Labor is by far the biggest warehouse cost (~60–65% of fulfillment expense), so even small inefficiencies (extra walking, rework) greatly inflate operating costs. Picking and stocking **errors** also eat profit: industry studies report 1–3% error rates, each error costing \$50–\$300 and roughly 11–13% of profitability.

Current solutions: Classical warehouse management systems (WMS) use rules-based slotting (e.g. ABC analysis, batch picking zones) and heuristic optimization (genetic algorithms, simulated annealing) to improve flow. Many firms implement conveyor and robotic automation. These tools offer gains: for instance, implementing/upgrading a WMS often yields ~10–15% cost savings. Nonetheless, warehouses still face combinatorial complexity. The **slotting optimization** problem is NP-hard – assigning n items to m locations has exponentially many configurations. State-of-art solvers (integer programming, local-search heuristics) can handle moderate instances, but large fulfillment centers (hundreds of thousands of SKUs and slots) push them to limits. Furthermore, static optimizers may not adapt quickly to changing demand or seasonality.

KPIs and costs: Warehouse inefficiencies impact cycle time, throughput, and service. Key metrics include: order picking speed (orders or lines per hour), labor utilization, dwell time (time inventory waits to be picked), and space utilization. Wasted moves and errors directly hit cost-per-order and throughput. For example, an internal case found that re-optimizing rack assignments via quantum-simulated methods could reduce re-handling by **10×**, sharply cutting labor. On the cost side, one industrial lab noted each minute of delay for an item can cost *thou-*

sands of dollars per day, and a single bad day in high volume can cost *millions*. More broadly, as of 2024 U.S. logistics spend exceeded \$2.58 trillion – even tiny percentage improvements in warehousing translate into very large savings.

Computational bottlenecks: Optimal slotting/layout planning can involve solving massive graph or flow problems. For example, fitting products to gravity-flow shelves with capacity and co-location constraints leads to a huge Quadratic Unconstrained Binary Optimization (QUBO) problem. Classical methods often resort to approximations or decomposition. Running simulations to test a layout (e.g. simulating order flow for months of operation) is also time-consuming. Updating the optimization in real time or near-real time (e.g. daily) is often infeasible.

Summary of challenges: Warehouse slotting and storage optimization remain largely manual or heuristic in many firms, with significant scope to improve. Current methods, while mature, still leave inefficiencies worth millions on the table (given high labor costs). Success metrics here include reduced travel time per order, higher picking rates, lower error rates, and improved labor productivity. Any technology that meaningfully accelerates the search for optimal layouts or rapidly re-optimizes in response to demand changes would have high ROI.

1.3 Combined Demand–Warehouse Optimization Problem

It is natural to integrate demand forecasting and storage optimization into a **single, joint problem**: *given predicted demand across products and time, simultaneously decide inventory allocations and shelf slotting across the network to minimize total cost and meet service targets*. In essence, this is a demand-driven, multi-echelon inventory-slotting optimization. Demand forecasts determine how much of each SKU should be stocked and where, while the resulting stock locations feed back into fulfillment efficiency and capacity constraints.

Interactions and complexity: Linking the two creates a much larger problem. For example, if forecasts for 10,000 SKUs across 20 locations are considered, the optimizer must decide how to distribute millions of units spatially and temporally. Errors in forecasting directly cause inefficiencies in allocation (under-forecast → stockouts at some nodes; over-forecast → excess inventory and wasted space). Conversely, warehouse constraints (limited slots, varying pick frequencies) can influence how forecasts should translate into inventory levels (e.g. batch picking might favor bulk stocking of fast-movers). Optimizing both layers simultaneously is a high-dimensional combinatorial problem: it combines stochastic demand modeling with NP-hard inventory-placement and slotting optimization. In computational terms, casting it all into one QUBO would be *extremely* complex. For context, even a pure slotting QAOA problem is QCMA-hard at optimal depth (quantum analog of NP-hard), so adding forecast uncertainty likely does not simplify the math.

Why quantum might help: Despite its daunting scale, quantum computing offers some attractions for this joint problem. First, quantum algorithms (like variational QNNs) might forecast demand patterns using fewer parameters and faster convergence than classical ML. Second, the inventory-slotting piece is a natural fit for quantum optimization (QUBO/QAOA, quantum annealing) which can natively handle binary allocation decisions. In principle, a single quantum-assisted solver could explore many combined “what-if” scenarios in superposition, effectively parallelizing scenario analysis. For example, QAOA circuits can encode both assignment costs and demand penalties. In early tests, hybrid QAOA has matched or outperformed classical heuristics on small inventory-allocation benchmarks. More broadly, industry analysts note that even a 1–2% improvement in logistics efficiency (achievable via quantum-inspired optimization) can yield multi-million-dollar savings for large operations.

Scalability and hardware maturity: Currently, quantum devices are limited (hundreds to thousands of qubits, high error rates). Fully solving a real-world joint planning problem is beyond to-

day's hardware. However, hybrid schemes can partially tackle subproblems: e.g. use quantum methods to slot a subset of high-volume SKUs or to refine classical solutions. Research prototypes have already demonstrated feasibility: for instance, one group implemented a gravity-rack slotting QAOA on trapped-ion hardware. Future quantum systems (thousands of robust qubits) could handle much larger instances. Meanwhile, major logistics firms (DHL, Maersk) and consortiums (QED-C) are actively exploring high-impact use cases in this space, suggesting enterprise interest is high despite nascent tech.

Combined problem statement: *Joint demand-warehouse optimization can be defined as:*
> Given historical data and current conditions, forecast future demand for each product. Then allocate inventory and choose shelf-slot assignments across warehouses to minimize total cost (transportation, holding, picking) subject to service constraints. The objective accounts for both forecast accuracy (penalizing mismatches) and warehouse operations (minimizing handling/replenishment).

This integrated problem inherently couples stochastic forecasting with combinatorial placement. It is scalable (grows exponentially with SKUs and locations) and naturally NP-hard. It thus stands as a candidate for QC: if quantum algorithms can deliver even modest speedups or solution-quality improvements, the financial impact justifies exploration.

1.4 Classical vs Quantum Approaches (Summary Table)

Problem	Classical Approach	Quantum Approach	Complexity	Maturity	Data Requirements
Demand Forecasting	ARIMA, exponential smoothing, ML models (LSTM, XGBoost, ensembles) using historical sales and market data.	Quantum Neural Networks, quantum-enhanced regressors (variational circuits).	Classical training is polynomial in data size; forecasts for large SKU sets can be very heavy on compute (especially for many features). Quantum training (hybrid) may be expensive per run but can reduce parameter count.	Classical ML forecasting is mature and widely deployed. Quantum forecasting (QML) is early-stage, limited to research pilots.	Large historical demand datasets, promotions, lead times, and external features. Quantum models require encoding classical features into qubit states (e.g. via angle encoding).

Problem	Classical Approach	Quantum Approach	Complexity	Maturity	Data Requirements
Warehouse Optimization	Mixed-Integer Programming, constraint programming, and heuristics (genetic algorithms, greedy slotting) within modern WMS.	Quantum annealing / QAOA on a QUBO formulation of slotting.	Both are NP-hard (combinatorial explosion). Classical solvers use heuristics (polynomial per iteration but many iterations needed). QAOA/QUBO is also NP-hard in worst case; optimal parameter tuning is QCMA-hard.	Classical optimization for warehousing (CPLEX, heuristic tools) is production-grade. Quantum optimization is experimental – small proofs-of-concept on NISQ hardware.	Warehouse layout data: rack positions/capacities, SKU sizes and co-location affinities. QUBO formulation requires a cost/affinity matrix between items and slots.

1.5 Prioritized Pain Points

1. **Forecast Inaccuracy:** High forecast error (20–40%) leads to huge stockouts/overstocks (>\$1T losses). Fragmented data and delayed planning cycles cause misalignment between demand and supply.
2. **Slow Model Refresh:** Long update cycles (weeks/months) cannot keep pace with daily demand shifts. This latency results in stale plans and missed opportunities.
3. **Warehouse Slotting Inefficiency:** Suboptimal item placement causes excessive travel and handling. Re-handling (e.g. gravity rack reinsertion) was **10×** higher in a naive layout, versus quantum-optimized layouts.
4. **Picking Errors and Delays:** Manual picking errors (~1–3% orders) impose a large cost (\$50–\$300/error). These degrade service and increase labor overhead.
5. **Scalability of Solutions:** Classical solvers struggle as SKU count and warehouse size grow. Even a few percent improvement at scale (1–2%) yields multi-million-dollar savings, but current methods plateau.

(These pain points apply across company sizes: small firms suffer from lack of data/tools, while large global carriers deal with sheer scale and complexity. Across industries – retail to manufacturing – demand volatility and inventory costs are common pains.)

1.6 Combined Optimization: Why It Matters

Integrating forecasting with warehouse planning is justified by the strong feedback loop: better forecasts enable smarter inventory placement, and optimized layouts improve responsiveness

to demand. For example, if forecast sees a surge in Product X, the system can proactively place X in high-throughput slots near shipping docks. Conversely, if warehousing data shows certain SKUs never move together, demand planners may bundle forecasts or promotions differently. Solving the problems jointly (versus in silos) can unlock synergies – but only if a powerful optimizer can handle the full scope.

Quantum computing fits naturally into this integrated problem. It can handle data-driven prediction (via QML) and large-scale combinatorial search (via QAOA/annealing) in one framework. Even with current hardware limits, hybrid quantum-classical workflows could tackle parts of the problem (e.g. cluster SKUs by demand with QC, then slot classically, or vice versa). The expectation is not instant enterprise deployment, but a roadmap: start with proof-of-concepts, targeting the biggest bottlenecks. Already, companies are investing: a 2022 survey found **63% of logistics firms** exploring quantum projects. Given the enormous cost base (global supply-chain spend in the trillions), even a small quantum-enabled improvement could justify the investment.

1.7 Quantum vs Classical: Key Metrics and ROI

- **Market Opportunity:** Logistics is a multi-trillion-dollar industry (US logistics \$2.58T in 2024; 3PL sector \$1.6T by 2035). Within that, savings from improved forecasting and warehousing run into billions. Quantum applications here are expected to drive significant ROI: e.g. D-Wave customers report time/cost savings (Pattison Foods saw an 80% cut in scheduling time).
- **Economic Impact:** As noted, 1–2% efficiency gains (feasible via optimization) correspond to millions saved. Return thresholds for new tech in enterprise logistics are typically low single-digit percentages. For example, automating a warehouse task often must justify payback in <1–2 years, which is realistic if even a small fraction of labor cost is cut.
- **Case Studies:** Quantum trials (D-Wave with VW and Toyota, Von Braun Labs with an automotive warehouse) show early promise on slotting/scheduling. Though none have fully solved real-scale problems yet, these pilots validate the *idea* that quantum can add value in logistics. News: Volkswagen’s quantum bus-routing demo, DHL’s quantum trend radar, Maersk’s QC research all indicate strong momentum.

<!--## Interview Questions for Large-Scale Logistics Executives

- **Demand Forecasting Challenges:** What factors currently limit your forecast accuracy (data gaps, variability, tooling)? How large are the typical forecast errors (MAPE)?
- **Forecast Impact Metrics:** Which KPIs suffer most from forecast errors? (e.g. service level, out-of-stock rate, inventory turns, emergency freight costs)
- **Current Methods:** What forecasting methods and software do you use (statistical vs ML)? How long does training/inference take for your largest forecasts?
- **Warehouse Slotting Practices:** How do you determine product slotting and inventory placement? Do you use WMS algorithms or manual rules? What are the pain points (e.g. frequent re-slotting needed, long walking distances)?
- **Warehouse Performance:** What KPIs reflect your warehouse efficiency? (order fulfillment time, picks/hour, cost per pick, error rate) How far are you from industry best-practice?
- **Integration:** Are demand forecasts and inventory placement decisions currently made in an integrated way? If not, what would it take to align them?
- **Computational Limits:** Where do you hit computational limits (e.g. slow scenario analyses, model retraining)? How important is faster computation or real-time re-optimization?

- **Improvement Priorities:** Would a 1–2% improvement in forecasting or warehousing efficiency be worth investing in? What ROI or payback period would justify a new technology?
- **Quantum Readiness:** Are you aware of quantum computing in logistics? Have you experimented with any pilot projects or vendors? What concerns or hopes do you have about it?

1.8 Metrics to Measure Success

To evaluate any new solution, track both **operational** and **financial** metrics, for example:

- **Forecast Accuracy:** MAPE, WAPE, bias; improvements should translate to fewer stockouts and overstocks.
- **Service Level:** Fill rate (percent of orders met), on-time delivery rate; should rise as forecasts improve.
- **Inventory Efficiency:** Days of inventory, turns per year, carrying cost as % of sales; better forecasts and slotting should allow leaner stock.
- **Order Cycle Time:** Throughput per day, average pick time per order, orders-per-hour; warehouse optimizations should speed up fulfillment.
- **Error Rates:** Picking/put-away error percentage; should decline with better automation/slotting.
- **Labor Productivity:** Units/picks per labor hour; small gains here scale to big savings (labor is ~60% of cost).
- **Computation Time:** Model training and optimization runtime; quantum approaches should aim to cut these compared to classical (e.g. from hours to minutes).
- **Cost Metrics:** Total logistics cost (transport+warehouse) per unit shipped; reduce this through overall gains. ROI is high if even 1–2% is trimmed given multi-million-dollar budgets.